

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Evaluated production system health and **AI model performance** using **Splunk, CloudWatch, and Dynatrace**, analyzing logs and metrics to identify reliability issues and validate real-world system behavior against expected outputs.
- Designed and implemented **Python-based data pipelines and ETL workflows** for analytics, including **statistical validation** of data quality and anomaly detection across enterprise datasets.
- Developed **REST APIs** and analytical features in **Java Spring Boot**, writing well-documented code and conducting peer reviews to ensure technical accuracy and maintainability.
- Performed **statistical analysis and data-driven validation** using **SQL (PostgreSQL, MySQL, Oracle)** to support hypothesis testing, ad-hoc reporting, and schema optimization across production databases.
- Automated quantitative workflows through **Python scripting, shell scripting, and cron jobs**, reducing manual effort and improving reliability of scheduled analytical tasks.
- Collaborated in **Agile/Scrum ceremonies** and documented technical findings in **Confluence**, communicating complex technical reasoning to cross-functional teams with clarity and precision.

SKILLS

Programming & Scripting – Python, Java, Shell scripting, SQL.

Data Engineering & Analytics – ETL pipelines, data validation, statistical analysis, pandas, NumPy, PostgreSQL, MySQL, Oracle.

Machine Learning & Modeling – Predictive modeling, hypothesis testing, regression, classification, experimental design, A/B testing.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD, Git.

Monitoring & Observability – Splunk, Dynatrace, log analysis, SLA management, incident response.

Frameworks & Tools – Spring Boot, REST APIs, Flask, FastAPI, Jira, Linux.

Quantitative Methods – Time-series forecasting, optimization, statistical inference, scientific reasoning about data and models.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Production Data Quality & Anomaly Detection Pipeline (HCL Technologies)

- Built end-to-end **Python ETL pipeline** to validate enterprise Java/.NET application outputs against expected statistical distributions and business logic.

- Implemented **anomaly detection algorithms** using **pandas and NumPy** to flag data integrity issues in real-time, reducing false positives by **30%** through iterative threshold tuning.
- Automated **statistical validation reports** using **scheduled Python scripts** and cron jobs, enabling quick identification of model drift and data quality degradation.
- Documented analytical methodology and findings in **Confluence** for cross-functional review, ensuring transparency in validation logic and assumptions.

REST API Feature Development & Technical Analysis (HCL Technologies)

- Developed **Spring Boot REST APIs** for data-driven features, writing clean, well-documented code and conducting peer code reviews to validate technical correctness.
- Performed **SQL-based exploratory analysis** to understand data patterns and inform API design decisions, using **PostgreSQL and Oracle** for schema queries and performance testing.
- Monitored API reliability in production using **CloudWatch and Splunk**, analyzing performance metrics and logs to validate that outputs matched expected behavior under real-world conditions.
- Collaborated with team members in **Agile ceremonies** to communicate technical findings and trade-offs, ensuring alignment on quantitative validation criteria.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)